

Nikolai Ovchinnikov

Software Engineer — Low-Latency C++ / Trading Systems / High-Load Backends

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Summary

Software engineer (~10 years) specialising in low-latency C++ and high-load backends. Currently on HFT trading platforms in modern C++ (20/23/26); previously FPGA tooling at Guzik, petabyte-scale PII storage at Booking.com, and mail backends at Yandex.

Experience

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|---|--|----------------|
| Software Engineer, Trading Systems | Maven Securities (hybrid) | 2025 - Present |
| <ul style="list-style-type: none">Shipped CBOE C2 (US options) and complex/combo orders end-to-end (FPGA, execution, deploy); delivered delta-aware net exposure capping across the market-making and hedging stacksInstrumented trading systems with per-user state-change attribution and exposed the full instrument-params inventory to the trading desk UI; refactored the generic message-bus subscriber to drop locking on a latency-sensitive hot pathAuthored the firm's onboarding guide and rolled out shared C++ editor tooling (clang-format, VSCode/Cursor) adopted firm-wideBuilt a low-latency "create-and-send" path in API — collapses instrument creation and order submission into one request, cutting round-trips on new-listing flows | | |
| Software Engineer, Data Privacy | Booking.com (hybrid) | 2022 - 2025 |
| <ul style="list-style-type: none">Led migration of a critical PII-storage service (Spring Boot, Apache Cassandra, Kafka) from bare-metal to Kubernetes — 400K RPS at ~90% lower costDrove DMA compliance: RFC, implementation, rollout monitoring — zero client-facing disruption | | |
| Software Engineer, Mail | Yandex (hybrid) | 2021 - 2022 |
| <ul style="list-style-type: none">Built C++17/20 backend services on one of Europe's largest PostgreSQL installations; integrated AWS SDK for C++ with Boost.Asio coroutinesShipped archive/restore of user mail data — ~10% PostgreSQL storage saved | | |
| Software Engineer | Guzik Technical Enterprises (hybrid) | 2019 - 2021 |
| <ul style="list-style-type: none">Built software for high-throughput analog-to-digital converters end-to-end (low-level HW → UI); led the Python/CPython extension for the digitizer SDKRebuilt FPGA CI/CD — 2–3× faster pipelines with better failure diagnostics | | |
| Freelance Software Engineer | Self-employed (remote) | 2014 - 2019 |
| <ul style="list-style-type: none">100+ projects in PHP, JavaScript, and Delphi — from minor enhancements to full-scale products | | |

Education

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|---|---------------------|
| Novosibirsk State University | Novosibirsk, Russia |
| B.Sc. in Computer Science, GPA 4.5 / 5.0 | 2015 - 2019 |
| <ul style="list-style-type: none">Thesis (C++17 / Qt / OpenCV): stereo-camera calibration and 3D reconstruction | |

Skills

Programming Languages: 🇨🇧 C++ (20/23/26), 🐍 Python, ☕ Java

Tools & Systems: AI/LLM, Boost, PostgreSQL, Docker, Cython, Linux, concurrency, async I/O, profiling, production debugging, low-latency systems.

Collaboration: onboarding new joiners across teams, mentoring junior/mid engineers, technical writing.

Languages: English (proficient), Russian (native), Dutch (A2)

Side Interests: [Reverse engineering](#) & [CTF prize-winner](#); hobby electronics (STM32, ESP32, Arduino); kitesurfing, padel.